

A Verified Certificate for Graham’s Rearrangement Conjecture in $\mathbb{Z}_{29}$

ERDÖSFIRE prospective computation

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Abstract

We give a reproducible computational certificate establishing the Graham/Alspach sequenceability conjecture for the cyclic group $\mathbb{Z}_{29}$. A recent general theorem [1] covers all subsets of cardinality at most 20. For every subset cardinality from 21 through 28, we quotient by the action of $\mathbb{Z}_{29}^\times$, produce a sequencing witness for each of 60,134 canonical orbit representatives, and independently verify exact coverage of all 1,683,218 target subsets. Two independently implemented verifiers use different coverage arguments: full orbit expansion into a 2^{28} -element bit universe, and canonical-orbit accounting by stabilizer size. The certificate is deterministic across thread counts and survives adversarial mutation tests.

1 Statement

For a finite subset A of an abelian group, call an ordering $a_1, \dots, a_m$ a sequencing when the partial sums $s_i = \sum_{j=1}^i a_j$ are pairwise distinct and $s_i \neq 0$ for $1 \leq i < m$.

Theorem 1. *Every subset $A \subseteq \mathbb{Z}_{29} \setminus \{0\}$ is sequenceable.*

2 Reduction by symmetry

Let $U = \mathbb{Z}_{29}^\times$ act on subsets by $u \cdot A = \{ua : a \in A\}$.

Lemma 1. *If $(a_1, \dots, a_m)$ sequences A , then $(ua_1, \dots, ua_m)$ sequences uA for every $u \in U$.*

Proof. Every partial sum is multiplied by the nonzero scalar u . Thus equality or vanishing of scaled partial sums is equivalent to equality or vanishing before scaling. $\square$

It therefore suffices to exhibit one sequencing for each orbit. We encode a subset of $\{1, \dots, 28\}$ by a 28-bit mask and choose the numerically least mask in each orbit as its canonical representative.

3 Certificate

The certificate contains one row per canonical representative for cardinalities 21 through 28. Each row stores the mask, cardinality, subset sum modulo 29, and an explicit ordering. Its SHA-256 digest is

4dd6b2b21f75ffdd1ff472d1bf9a70757ccea0b31f147d49c6b1ff675c5ef961.

$ A $	all subsets	orbit representatives
21	1,184,040	42,288
22	376,740	13,468
23	98,280	3,510
24	20,475	735
25	3,276	117
26	378	14
27	28	1
28	1	1
Total	1,683,218	60,134

Table 1: Certified search domain and multiplicative-orbit reduction.

4 Independent verification

The Go verifier parses the certificate independently, validates each representative ordering, scales each witness by all 28 units, and marks the resulting subsets in a bitset indexed by all 28-bit masks. It checks exact cardinality counts and reports 1,683,218 unique covered subsets after 1,683,752 scaled-witness checks.

The Python verifier independently recomputes each canonical orbit and its stabilizer, validates every representative witness, rejects duplicate canonical representatives, and sums the exact disjoint orbit sizes. For each cardinality k , the sum equals $\binom{28}{k}$.

The two methods agree on every cardinality, the total, the representative count, the orbit-size histogram, and the certificate digest. Both reject certificates with a missing representative, a duplicated orbit, a corrupted ordering, or a false total field.

5 Proof of the theorem

For $|A| \leq 20$, invoke the published general small-set theorem [1]. For $21 \leq |A| \leq 28$, exact coverage verification places A in the orbit of a unique canonical representative R in the certificate. The certificate supplies a verified sequencing of R ; the symmetry lemma scales it to a sequencing of A . These ranges exhaust all subsets of $\mathbb{Z}_{29} \setminus \{0\}$.

6 Reproducibility and scope

The witness generator is not trusted as a verifier. A second generation with a different thread count produced the identical byte-level certificate. The complete sources, certificate, verification reports, and mutation tests accompany this manuscript. The mathematical claim is computationally verified; external peer review and priority adjudication are not asserted here.

References

- [1] S. Costa, S. Della Fiore, M. Fontana, and L. Vena, *Graham conjecture on small sets in abelian groups*, arXiv:2603.20961 [math.CO], 2026.